Chapter 4

Post-CMP Cleaning

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Chapter Contents

1 INTRODUCTION

With the rapid shrinking of the device dimensions and the strict requirements of achieving extremely smooth surfaces, there is an increasing need for global surface planarization of various thin films in the front- and back-end processes using chemical mechanical planarization (CMP) [1]. During CMP, the pad material and the wafer surface are in intimate contact in the presence of slurry [2]. The slurry particles remain on the wafer surface at the end of the CMP process, and if not removed, they can cause various types of defects (scratches, corrosion spots, etc.) in the subsequent processing steps, which affects the functionalization of the integrated circuit (IC). These defects, in some instances, may also arise from the pad and the diamond disc conditioner. The other common forms of contaminants include organic residues and metallic impurities. The residues typically originate from slurry additives or the pad material and require subsequent processing such as plasma ashing or an oxidant (e.g., peroxide)-based chemical cleaning [3]. The metallic impurities that are left on the surface in concentrations of 10^{11}-10^{12} atoms/cm^2 emanate either from

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TABLE 4.1 Contamination and Defects in CMP

Contaminants	Nonmetal CMP	Metal CMP	Effects
Particulate	Silica or ceria, fine fragments of film or pad	Alumina, or silica, metal hydroxide precipitates., fine fragments of film or pad	(1) Cause local roughness and block photolithography (2) Pinholes in new grown films: metal precipitates leads to metallic contamination (3) Shorts by conductive particles
Metallic	K^+, Ca^{2+}	$W_xO_y^{(2y-6x)-}$, Cu^{2+}, Al^{3+}, Fe^{3+}, IO_4^-, I^-/I_2, $Fe(CN)_6{}^{3-}$, $Fe(CN)_6{}^{4-}$	(1) Alkali metal ions: high mobility influences electrical characteristics (2) Copper: fast diffuser in Si (3) Many metals can form silicide and/or affect the oxidation (4) Noble metal ions cause etching of Si
Organic	Tetramethyl ammonium salts, buffers, surfactants	Buffers, surfactants, inhibitors	(1) Affects wettability and cleanability (2) Out gassing (3) Poor adhesion of deposited layers
Other defects	Scratches, stress	Scratches, puddling, plug coring, dishing and erosion, stress	(1) Nonplanar surface (2) Filling of other material (3) Potentially causing stringers (4) Weak sites

Reprinted from Ref. [4].

abrasion caused to the metal lines or from metal ions present in the slurry. Table 4.1 summarizes different types of contaminants and their effects during processing of devices [4]. Particulate contamination in the form of abrasive particles, precipitates, or fragments of films and pad can enhance local surface roughness, impact the photolithography process by blocking the UV light, or cause shorting when the particles are conductive. Metallic impurities can affect electrical characteristics when they are highly mobile or cause dissolution of silicon in the case of more noble metal ions. Organic residues influence the wettability of the films and reduce the adhesion of deposited films.

In an effort to significantly lower the defect density on various films prior to the next processing step, the demand for an effective and efficient post-CMP cleaning process has been continuously rising. As a result, there is a tremendous interest towards improving the fundamental understanding of the interfacial phenomena and the underlying physical and mechanical mechanisms that drive the cleaning process with the ultimate objective of developing innovative approaches with newer chemistries and advanced tools that can effectively achieve desired cleaning with minimum damage. In this chapter, a review of batch- and single-wafer brush and megasonic cleaning processes with emphasis on the effect of different process variables on particle removal is provided. The first few sections briefly describe the forces that drive particle adhesion and removal in a post-CMP cleaning process. The last part of the chapter focuses on the typical chemical formulations and the role of additives in post-CMP cleaning of silicon dioxide, tungsten, and copper films.

2 FORCES ON PARTICULATE CONTAMINANTS IN A POST-CMP CLEANING PROCESS

A particle on a polished surface immersed in a liquid and subjected to an external force such as that in a brush cleaning or megasonic cleaning experiences several types of forces acting on it. Some of these forces and their dependency on the particle dimension are shown in Table 4.2 [5]. As can be seen from this table, the dominant forces are van der Waals adhesion forces and double-layer interactions (electrostatic forces of interaction) due to their higher magnitude compared to other forces for the same size of the particle.

2.1 van der Waals Forces

The van der Waals forces are one of the most important forces in the adhesion of particles to any substrate. These forces originate due to the instantaneous dipole interactions between molecules formed due to the positions of the electrons surrounding the nuclei. There are three types of dipole interactions, namely, permanent dipole-permanent dipole (Keesom interactions), permanent dipole-induced dipole (Debye interactions), and induced dipole-induced dipole interactions (London forces) which collectively contribute to van der Waals forces [6]. The van der Waals forces depend on the composition of the interacting particle, substrate and the medium, the distance of separation between them, and the surface roughness and geometry of the particle and the substrate [7]. When the microscopic dipole interactions are summed over all the molecules in a macroscopic body, van der Waals forces between different geometries

TABLE 4.2 Rough Magnitudes of Various Forces on Particles (Silicon Nitride) of Various Dimensions in Contact with a Flat Surface Immersed in DI Water at Room Temperature (Surface Potential of Substrate Considered ~−300 mV)

Nature of Forces	Particle Dimension Dependencies	Forces, Order of Magnitude (*N*)		
		R = *0.1 μm*	R = *1 μm*	R = *10 μm*
van der Waals	*R*	10^{-7}	10^{-6}	10^{-5}
Electrostatic	—	10^{-7}	10^{-6}	10^{-5}
Surface tension	*R*	10^{-8}	10^{-7}	10^{-6}
Drag	*R*	10^{-9}	10^{-8}	10^{-7}
Gravitation	R^3	10^{-16}	10^{-13}	10^{-10}
Archimedes	R^3	10^{-17}	10^{-14}	10^{-11}

Reprinted from Ref. [5].

can be obtained. For example, the van der Waals forces between a spherical particle of radius R and a flat surface are given by Eq. (1)

$$F_{\mathrm{vdW}} = \frac{A_{132}R}{6Z^2} \tag{1}$$

where R is the radius of the particle, A_{132} is the Hamaker constant for particle 1 and substrate 2 with medium 3 in between, and Z is the distance of separation between the particle and the substrate.

The potential energy of interaction, obtained by integrating Eq. (2) over the distance of separation, is given by

$$E_{\mathrm{vdW}} = -\frac{A_{132}R}{6Z} \tag{2}$$

The Hamaker constant A_{132} can be evaluated using the equation

$$A_{132} = A_{33} + A_{21} - A_{32} - A_{31} \tag{3}$$

where A_{ij} is the Hamaker constant for two particles in vacuum.

A geometrical mixing rule relates the Hamaker constants between similar and dissimilar particles. Table 4.3 presents Hamaker constants for some materials of practical interest [8,9]

$$A_{ij} = \left(A_{ii}A_{jj}\right)^{1/2} \tag{4}$$

In the case of soft particles, especially when they are aged on the wafer surface for sufficient time, deformation can occur which can increase the contact area between the particle and the surface, leading to enhanced adhesion forces. Depending on the nature of deformation, such as elastic or plastic, different theories including Johnson-Kendall-Roberts, Derjaguin-Muller-Toporov, and Maugis-Pollock have been proposed for computing the contact area between the particle and the surface [10–12]. Kumar and Beaudoin have shown that the van der Waals force for a deformed smooth particle and a substrate can be calculated using Eq. (5) [7]

$$F_{\mathrm{vdW}} = \frac{A_{132} d}{12h^2}\left[1 + \frac{2a^2}{hd}\right] \tag{5}$$

where h is the distance of separation between the particle and the surface, a is the contact radius, and d is the diameter of the particle.

When surface roughness is taken into account, a Rumpf equation [13,14] can be used which considers the asperities to be hemispherical on the substrate

$$F_{\mathrm{vdW}} = \frac{A_{132} R}{6a^2}\left[\frac{1}{1 + \frac{R}{r}} + \frac{1}{\left(1 + \frac{r}{a}\right)^2}\right] \tag{6}$$

TABLE 4.3 Hamaker Constants for Different Materials [6,8,9]

Materials	Hamaker Constant (J) in Vacuum ($A \times 10^{20}$ J)
Alumina	15.4
Gold	45.3
Silver	39.8
Polystyrene	7.8-9.8
Metals	16-45
Si_3N_4	16.7
β-Si_3N_4	18
SiO_2	5.5-6.3
TiO_2	15.3
KCl	5.5
NaCl	6.5
CsI	8.0
Toluene	5.4
Acetone	4.2
Water	4.4

where R is the radius of the particle, r is the radius of the asperity, and a is the distance of separation between the particle and the asperity.

2.2 Double-Layer Interactions

When a substrate is immersed in a solution containing ions, it can acquire charges either by adsorption of specific ions or by dissociation. The potential of the charged surface depends on the activity of the potential-determining ions and is given by the Nernst equation shown below:

$$\Psi_0 = \frac{2.303RT}{F} \log\left(\frac{a}{a_{\mathrm{IEP}}}\right) \tag{7}$$

where ψ_0 is the potential difference between the charged surface and the bulk solution, a and a_{IEP} are the activities of the potential-determining ions in the solution and isoelectric (ISE) point, respectively, R is the gas constant, T is the temperature of the system, and F is the Faraday's constant.

The charging of a surface affects the distribution of ions surrounding it, resulting in an increased concentration of counterions and a decreased concentration of co-ions. The layer of liquid in immediate contact with the solid surface is strongly bound and is referred to as the Stern layer. The ions surrounding the Stern layer are mobile, and the concentration profile of ions in the region is a result of a balance between the thermal energy of the ions and the electrostatic interaction between the ions and the charged surface. This region is referred to as the diffuse layer. The Stern layer and the diffuse layer collectively form a double layer. A shear plane exists in the double layer that is described as an imaginary sphere around a particle inside which the solvent moves with the particle as the particles move through the solution. The potential at the shear plane is known as the zeta potential (Z_P). A schematic of different potentials associated with colloids in a solution is shown in Fig. 4.1 [15].

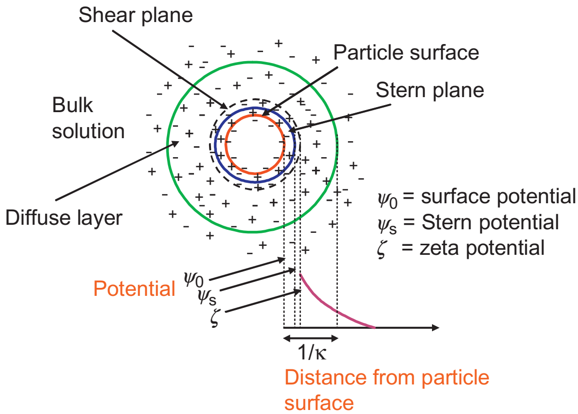

FIGURE 4.1 Schematic representation of different potentials associated with a particle in aqueous solution. *(Reprinted from Ref. [15].)*

Assuming that the surface potential is less than 25 mV, also known as the Debye-Hückel approximation, Eqs. (8) and (9) give the variation of potential as a function of distance from a flat and spherical surface, respectively [6].

$$\Psi = \Psi_0 \exp(-\kappa x) \tag{8}$$

$$\psi = \psi_0 \left(\frac{a}{r}\right) \exp[-\kappa(r-a)] \tag{9}$$

where ψ is the potential at a distance x or r from the surface, ψ_0 is the surface potential, a is the radius of the particle, and κ is the inverse Debye length given by Eq. (10)

$$\kappa = \left[\left(\frac{2000e^2N}{\varepsilon k_B T}\right) I\right]^{1/2} \tag{10}$$

where e is the electronic charge, N is the Avogadro's number, k_B is the Boltzmann constant, and I is the ionic strength of the solution.

$\frac{1}{\kappa}$, an important parameter, gives the distance at which the potential drops to $\frac{\Psi_0}{e}$ and is often referred to as the double-layer thickness.

Both surface potential and Debye length are important parameters in determining the electrostatic energy of interaction between charged surfaces. Such energies of interaction between a charged sphere and a flat surface with surface potentials Ψ_{01} and Ψ_{02}, respectively, are given by Eq. (11) [6]

$$W_E = \pi \varepsilon a \left({\Psi_{01}}^2 + {\Psi_{02}}^2\right) \left\{ \frac{2\Psi_{01}\Psi_{02}}{{\Psi_{01}}^2 + {\Psi_{02}}^2} \ln \frac{1+\exp(-\kappa D)}{1-\exp(-\kappa D)} + \ln[1-\exp(-2\kappa D)] \right\} \tag{11}$$

where W_E is the electrostatic energy of interaction between the sphere and the surface, D is the shortest distance between the flat surface and the spherical surface, a is radius of the sphere, and κ is the inverse Debye length.

During post-CMP cleaning, depending on the pH and ionic strength of the solution and use of specific additives, the zeta potential of the particles and the substrate will be affected and will influence the cleaning efficiency as will be discussed in the later sections.

3 TYPES OF POST-CMP CLEANING PROCESSES

3.1 Batch Cleaning

Among different types of wet cleaning methods, batch cleaning has been used for a long time due to its low cost of ownership (COO) as it usually processes a cassette of 25 wafers at a time. The wet batch cleaning is composed of several steps designated with different immersion tanks containing different chemicals. The technique includes immersion of wafers in a sequence in different cleaning solutions contained in separate baths [16]. This method of cleaning allows uniform contact of chemical solution on both sides of the wafers [17].

In addition, a good control of solution temperature can be obtained using this technique. In a typical wet bench used in industry, the wafer cassette is immersed in a proper cleaning bath maintained at a desired temperature for a specified process time followed by rinsing in DI water in another tank. The wafer may be immersed again in another chemical bath if required and the sequence may then be repeated. This technique, though very useful, requires significant amounts of chemicals and water. Metallic impurities, organic contaminants, and soluble material can easily be removed from the wafer surface using strong chemistries, but removal of particulate contamination often requires mechanical agitation such as megasonics and bubbling inert gas through the solution [16]. These impurities, including some from the wafer cassettes themselves, accumulate in the bath with subsequent cleaning, and hence bath replacement or recirculation using filters is required. In order to reduce contamination due to the wafer cassettes, silicon or carbide carriers are used to reduce the extent of contact with the wafers.

Manual wet benches have been replaced by automatic ones which reduce contamination and yield better results. A schematic of immersion cleaning is shown in Fig. 4.2.

In the first step, wafers are usually placed into an immersion tank containing sulfuric-peroxide mixture for organic contamination removal. According to different process requirements and immersion times, the ratio of H_2SO_4 (98%) to H_2O_2 (30%) is varied from $4H_2SO_4$:$1H_2O_2$ to $2H_2SO_4$:$1H_2O_2$ at temperatures of 120 °C or higher. Mixing of sulfuric acid and hydrogen peroxide solution is an exothermic process with final solution temperature depending on the composition and initial temperature of solution. The powerful oxidants, HSO_5^- and $H_3O_2^+$, formed during the reaction between H_2O_2 and H_2SO_4 can oxidize the organic impurities on the wafer surface [18–20]. SPM treatment is usually followed by a quick dump rinse process, which is commonly assisted by megasonic irradiation at a very high DI water flow rate.

SC-1 cleaning involves using mixtures of ammonium hydroxide (29%), hydrogen peroxide (30%), and water in the ratios from $1NH_4OH$:$1H_2O_2$:$5H_2O$

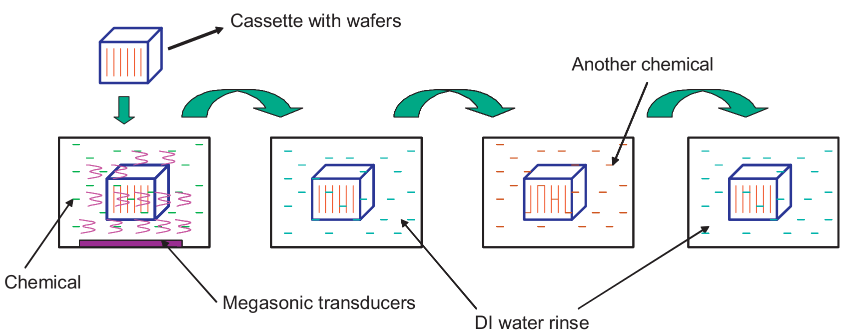

FIGURE 4.2 Schematic showing immersion cleaning of wafers.

to $1NH_4OH{:}1H_2O_2{:}200H_2O$ at 70-80 °C. The pH of this solution is close to 10-11. The use of temperatures higher than 80 °C is not advised in SC-1 solutions due to the rapid decomposition of peroxide and loss of NH_4OH as NH_3. At the alkaline pH, formation of HO_2^- species occurs. HO_2^- is a strong oxidant with the potential to remove organic contaminants from the wafer surface [21,22]. This species can also oxidize silicon to silicon dioxide which slowly etches in an alkaline solution. The continuous formation and etching of oxide aid in the removal of particles from the wafer surface. The SC-1 solution is also found to be useful in complexing metal ions such as Ag^+ and Cu^{2+} and in the dissolution of metals (silver, copper, chromium, cobalt, etc.).

After particle contamination removal, an SC-2 (standard clean-2) tank is used to remove metallic contaminants by forming complexes with Cl^- ions. The SC-2 cleaning solution consists of a mixture of hydrochloric acid (37%), hydrogen peroxide (30%), and water in a ratio of $1HCl{:}1H_2O_2{:}5H_2O$ to $1HCl{:}1H_2O_2{:}200H_2O$ at 70-80 °C. The pH of this solution is less than 1. Metal ions such as Fe^{3+}, Al^{3+}, Zn^{2+}, and Mg^{2+} hydrolyze in the SC-1 solution and form insoluble metal hydroxides that are not removed. Hence, the SC-2 solution is used since it can dissolve any metal as a metal ion due to the acidic nature of the solution. Although oxidation of silicon is possible in SC-2 due to the formation of powerful oxidants (e.g., $H_3O_2^+$) from hydrogen peroxide, there is no etching of either silicon or silicon dioxide.

After SPM, SC-1, and SC-2 treatments for post-CMP cleaning, a thin oxide layer is formed due to the oxidizing nature of the cleaning chemistry. Dilute HF (DHF) solution is used to remove the oxide layer from wafer surface, which additionally removes the metallic contamination adsorbed or trapped within the oxide layer. The etching rate and uniformity of HF solutions depend on the composition and temperature of the solution. Typically, HF solutions are used in the dilute concentrations of about 1HF (49%):$100H_2O$ at 25 °C with oxide (thermal oxide) etch rate of ~2 nm/min. Etching by HF leaves the surface terminated with —SiH or —SiH_2 groups. The etching of silicon dioxide by HF occurs according to Eq. (12) [16,23,24]:

$$SiO_2 + 6HF = SiF_6^{2-} + 2H^+ + 2H_2O \tag{12}$$

The last step for batch cleaning is the drying process, which is usually performed by placing wafers in an isopropyl alcohol (IPA)-water mixture at high temperatures. It should be mentioned that after each chemical treatment step, a rinsing process with DI water is followed in a separate immersion tank.

There have been a few studies on the cleaning chemistry development or process optimization in batch post-CMP cleaning. Abbadie et al. tested different cleaning chemistries on polished reclaimed wafers using an automatic wet bench [25]. The full cleaning consisted of initial cleaning or post-CMP cleaning followed by final cleaning. The post-CMP cleaning sequence is

TABLE 4.4 Three Initial Cleaning Procedures

SC-1 Standard Cleaning	HF/SC-1 Diluted Chemistry	HF/O_3-Based Chemistry
	HF:DIW = 1:100	HF:HCl: DIW = 1:1:100
	5 min	5 min (pH 1)
	DI water rinse	DI water rinse: 0.01% HCl
	10 min	3 min + O_3 6 ppm 7 min
SC-1 at 60 °C	SC-1 at 65 °C	
(NH_4OH:H_2O_2: H_2O = 1:1:6) 10 min	(NH_4OH:H_2O_2: H_2O = 1:4:80) 5 min	
DI water rinse	DI water rinse 50 °C	
10 min	8 min	
Centrifugation dry	Dry IPA vapor	Dry isopropyl alcohol vapor

Reprinted from Ref. [25].

shown in Table 4.4. Chemical concentration and cleaning time for each step have been illustrated.

By introducing ozone and acidic solution into the cleaning process, the authors developed two more cleaning procedures as illustrated in Table 4.5.

Particle removal, organic removal, metal removal, and environmental benefits were evaluated using the three cleaning processes. The results showed that during wet batch cleaning, HF-ozone process allows to achieve a wafer surface protected by a chemical oxide, which is atomically flat with a reduced particle and metallic contamination level. Additionally, acidic conditions at the beginning of the cleaning process minimized the particle contamination.

Tseng et al. developed a hybrid cleaning process with batch cleaning assisted by megasonic combined with brush cleaning for back end of line post-copper CMP [26]. In their study, both acidic and basic chemistries were used. The levels of trace Al detected by total reflection X-ray fluorescence (TXRF) are shown in Table 4.6.

Different methods and substrates were used. The results indicated that such hybrid cleaning processes combining both acidic and basic cleaning in sequence are able to achieve a more than 60% reduction in all CMP-related defects. AlO_x abrasives on the wafer edge were also greatly reduced by the hybrid cleaning process.

TABLE 4.5 Three Final Cleaning Procedures

Conventional RCA Cleaning	LC FEOL	DDC
	O_3 rinse 6 ppm (7 min)	
SC-1 at 60 °C	SC-1 at 65 °C	HF:HCl:DIW = 1:1:100
(NH_4OH:H_2O_2:H_2O = 1:1:6) 10 min	(NH_4OH:H_2O_2:H_2O = 1:4:80) 10 min	1 min
DI water rinse	DI water rinse 50 °C	DI water rinse: 0.01% HCl
10 min	8 min	3 min + O_3 6 ppm 7 min
SC-2 at 60 °C		
(HCl:H_2O_2:H_2O = 1:1:6)		
10 min		
DI water rinse		
10 min		
HF:DIW (0.6 to 1:100)	HF:DIW (0.6 to 1:1:100)	HF:DIW (0.6 to 1:1:100)
1 min	1 min	A few seconds
Final DI water	DI water rinse: 0.01% HCl	DI water rinse: 0.01% HCl
	3 min + O_3 6 ppm 7 min	3 min + O_3 6 ppm 7 min
Centrifugation dry	Dry isopropyl alcohol vapor	Dry isopropyl alcohol vapor

LC FEOL, low consumption front end of line; DDC, diluted dynamic clean.
Reprinted from Ref. [25].

TABLE 4.6 The Levels of Trace Al Left on Different Substrates After Cleaning

Trace of Al by TXRF Analysis	Al Density (a.u.)	
Wafer and CMP Cleaning Process	*Center*	*Edge*
a. Blanket Cu, Hybrid	0.36	0.25
b. Blanket Cu	0.4	0.26
c. Blanket Cu, ref. (no CMP process)	0.38	0.33
d. 22 nm patterned wafer	1.23	100
e. 22 nm patterned wafer, Hybrid	0.48	5.46

Reprinted from Ref. [26].

3.2 Single Wafer Cleaning

3.2.1 Brush Cleaning

Brush scrubbing is used for hydrodynamic removal of particulate contaminants and organic films from the wafer surface by rotation of a nylon or polypropylene brush across the surface. In the scrubbing process, the rolling of the brush can generate drag, electrostatic double layer, and thermophoretic forces on the particle adhered on the wafer surface [27].

The relative motion between the brush and the wafer causes a thin liquid flow which generates a drag on the particle. The thin film, along with the hydrophilic nature of the brush surface, provides a cushioning effect between the surfaces and prevents any scratches or damage to the wafer surface from the bristles of the brush or a large particle. The moment of the drag force can cause rolling of the particle and its removal from the surface. Electrostatic double-layer forces arise due to the surface charge and double-layer charge interactions [28].

In the thermophoretic mechanism, temperature gradients are employed between the wafer and the brush which generate a flow field and force in the direction opposite to that of the adhesion force between the particle and the wafer [29]. A schematic illustrating the forces acting on the particle in brush scrubbing is shown in Fig. 4.3.

The common chemicals used in brush scrubbing include water, surfactants, ammonium hydroxide solution, IPA, methanol, and others. The organic solvents help to remove organic films and also speed up the drying process. Although brush scrubbing is limited by its capability to remove submicrometer size particles, it is still a frequently used technique for post-CMP cleaning [30–32]. Another limitation of the brush scrubbing technique is that it is unable to remove particles from patterned wafers. This is due to the large size of the bristles which cannot penetrate the thin space between the features. The design of the brush scrubber is very critical and is often considered an art. An improperly designed scrubber can easily get clogged with debris and damage the wafer

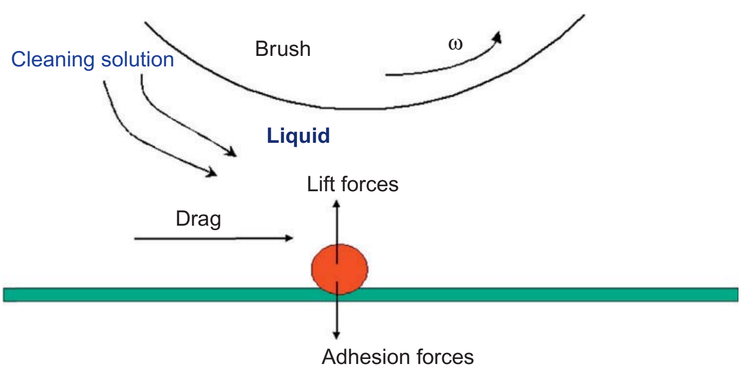

FIGURE 4.3 Schematic of brush scrubbing process. *(Reprinted from Ref. [27].)*

surface. Routine maintenance and replacement of scrubbers is another requirement associated with brush scrubbing.

The removal of particles during brush cleaning can occur by three mechanisms, namely, noncontact, partial contact, and full contact [33]. In noncontact mode, the spinning brush does not come in contact with the particle adhered on a wafer surface, as illustrated in Fig. 4.4a. The rotating brush causes the liquid layer between the particle and the wafer to move, which establishes a velocity gradient with maximum velocity of liquid at the surface of the brush and zero velocity at the wafer surface. The liquid velocity exerts a hydrodynamic drag force on the particle and causes its dislodgement from the wafer by rolling if the rolling moment (RM) (defined by Eq. (13)) is greater than 1 or removal moment is higher than the adhesion resisting moment. Partial-contact mode is a special type of noncontact mode where the distance between the top surface of the particle and the brush bottom surface is zero. The liquid velocity at the top of the particle is equal to that of the brush surface velocity. In full-contact mode, the brush is in surface contact with the particle and removes the particle by engulfing and creating a rotational torque. The removal moment, given by Eq. (14), is a function of power (I_b) and rotating speed (ω_b) of the brush. Scratching and defects on the wafer can be expected to occur in full-contact mode

$$\mathrm{RM} = \frac{F_d(1.399R - \delta) + F_{el} \cdot a}{F_a \cdot a} \tag{13}$$

where $\delta = R - \sqrt{R^2 - a^2}$

$$\mathrm{RM} = \frac{(I_b/\omega_b) + F_{el} \cdot a}{F_a \cdot a} \tag{14}$$

where R is the particle radius, a is the contact radius, F_d is the drag force, F_{el} is the electrostatic double-layer force, and F_a is the adhesion force.

Figure 4.5a and b shows plots of calculated RM versus bush rotating speed for silica particles of various sizes and at different distances from the brush surface for noncontact and full-contact modes, respectively. The calculations were performed for SC-1 solution taking into account the electrical double-layer force between the silica particle and the poly(vinyl alcohol) brush. As can be seen from Fig. 4.5a, the RM is a function of brush rotating speed and the distance between the particle (0.1 μm) and the brush (h). The RM is lower than 1 for distances higher than 5 μm irrespective of brush rotating speed, indicating that the particle will not be removed under these conditions. Only when the distance between the particle and the brush is 5 or 1 μm, the RM value is greater than 1 for brush speeds higher than 125 and 25 RPM, respectively. In this case, the hydrodynamic force typically becomes higher than the adhesion force and the particle is rolled away from the surface. In full-contact mode brush cleaning, when the particle is being engulfed, the brush is compressed, which increases the contact area between the particle and the brush and is about two times larger

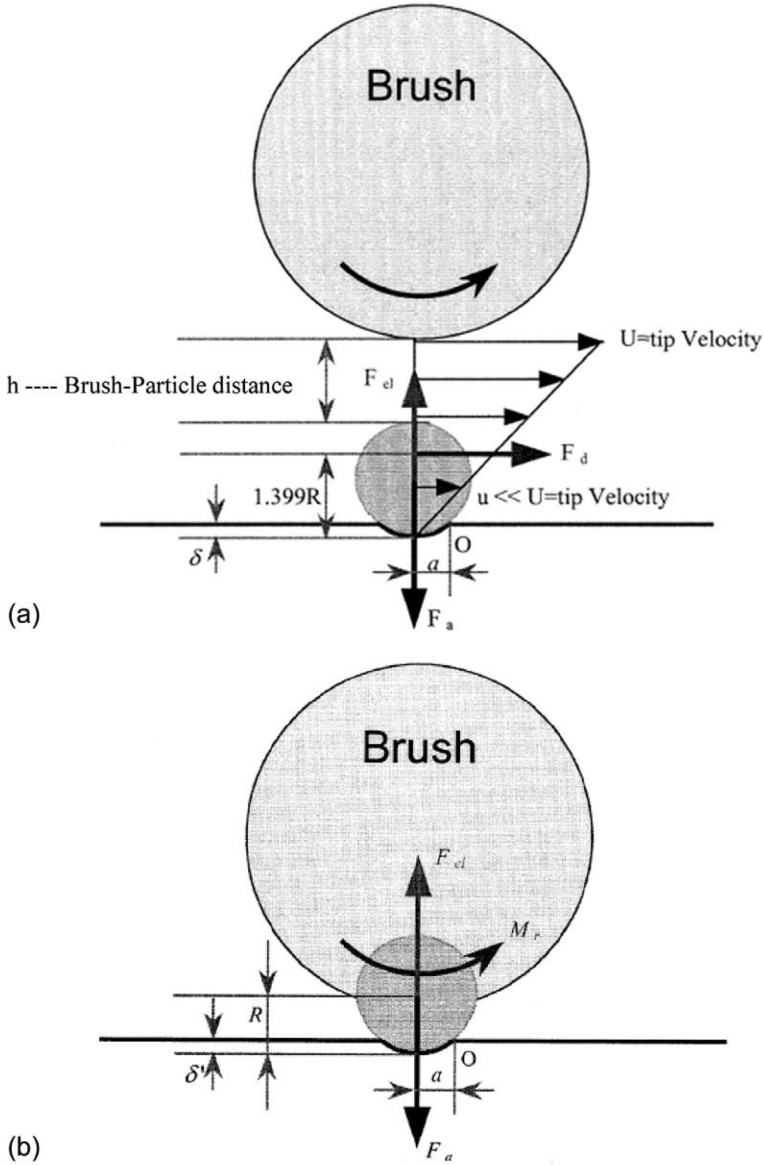

FIGURE 4.4 (a) Noncontact and (b) full-contact modes of particle removal mechanisms during brush cleaning. F_d is the drag force, F_{el} is the electrical double-layer force, F_a is the adhesion force, R is the radius of the particle, and a is the contact radius. *(Reprinted from Ref. [33].)*

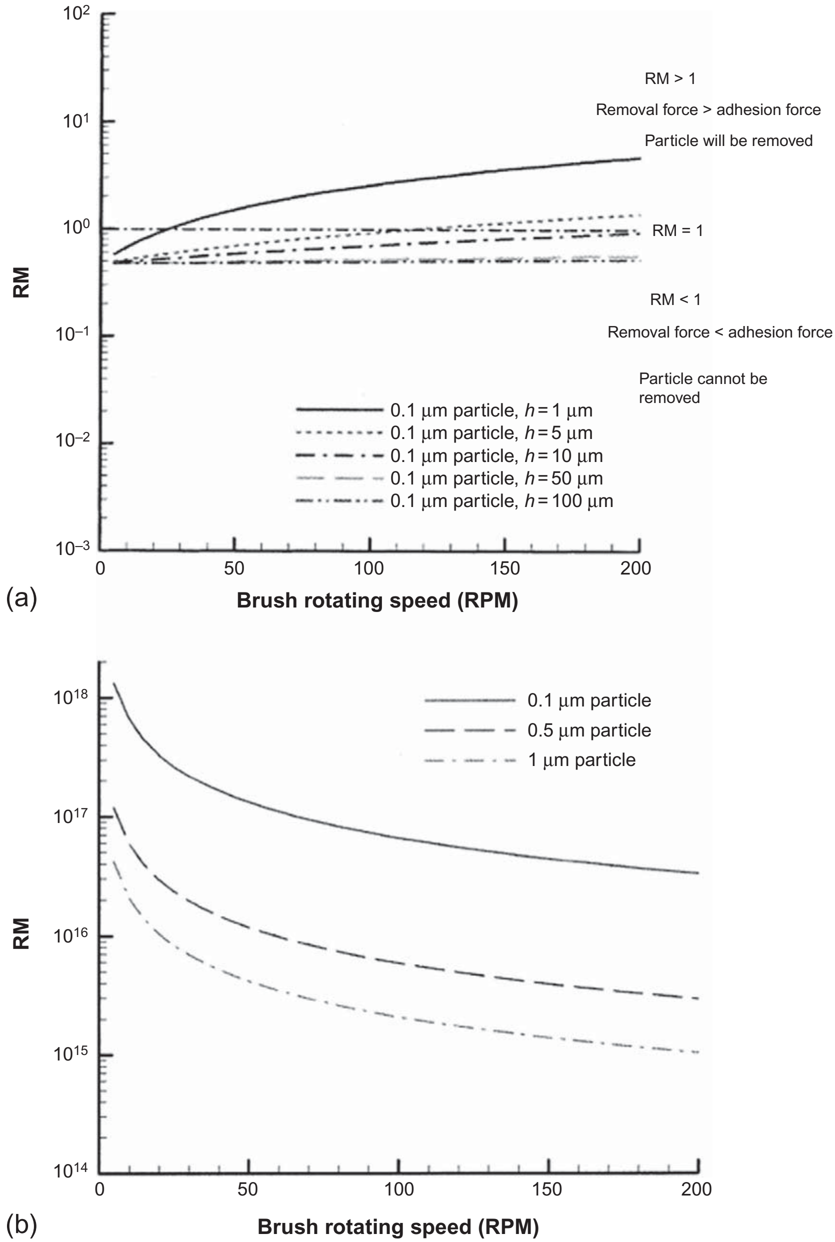

FIGURE 4.5 Effect of brush speed on RM (a) noncontact mode, 0.1-μm silica particle, SC-1 solution, (b) full-contact mode, 0.1-1.0 μm silica particles, SC-1 solution. *(Reprinted from Ref. [33].)*

than the contact area between the particle and the wafer surface. The particle adheres to the brush and is detached from the substrate due to rotational torque. When the particle is moved away from the surface, the brush is no longer compressed and the contact area between the particle and the brush reduces dramatically. The RM in full-contact mode is about 10^{15} or higher for silica particles of sizes between 0.1 and 1.0 μm and brush speeds lower than 200 RPM as illustrated in Fig. 4.5b. The RM is higher for smaller particles and lower speeds. The high values of RM greater than 10^{15} are likely to be sufficient to even remove silica particles bonded to the silicon dioxide surface by a hydrogen or a covalent bond.

Philipossian and Mustapha investigated the tribological attributes of post-CPM brush scrubbing of thermally grown silicon dioxide films on Si wafers as a function of applied pressure, solution pH, and flow rate of the cleaning solutions [34]. The solution pH was adjusted by addition of hydrochloric acid or ammonium hydroxide to DI water. The tribological characteristics were represented using the ratio of the shear force to normal force, also known as coefficient of friction (COF), while the brush velocity and the applied pressure were grouped in a dimensioless parameter defined as Sommerfeld number (So) as below:

$$\mathrm{So} = \frac{\mu \times U}{p \times h_{\mathrm{eff}}} \tag{15}$$

where μ is the viscosity of the cleaning liquid, U is the relative velocity between the brush and the wafer, p is the applied pressure to the wafer, and h_{eff} is the effective thickness of the liquid between the wafer and the brush.

Stribeck curves, which are plots of COF versus Sommerfeld number on a log-log scale, are good indicators of the lubrication mechanisms when two or more surfaces are in contact [35]. A typical Stribeck curve is shown in

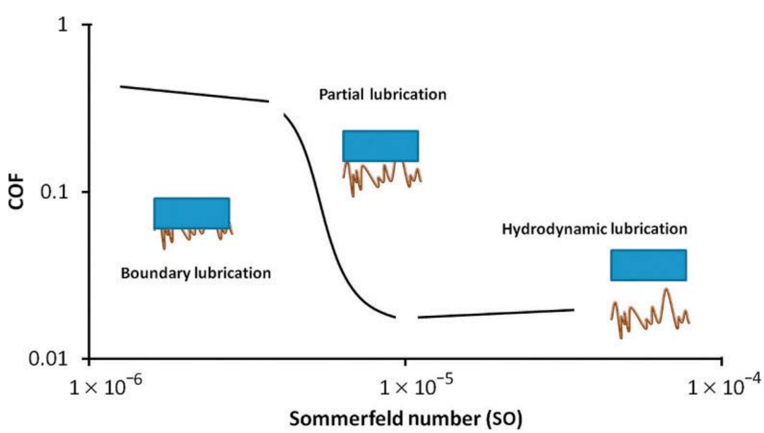

FIGURE 4.6 Generic Stribeck curve based on Sommerfeld number. *(Reprinted from Ref. [36].)*

Fig. 4.6 [36]. At low values of So, the wafer and the brush are in intimate contact with each other with little fluid between them. The COF values are relatively large and do not change much with increase of So. The mode of lubrication in this case is referred to as boundary lubrication. The second mode of lubrication, known as partial lubrication, occurs at intermediate values of So, when a thin film of liquid of thickness is roughly the same as the brush surface roughness exists between the wafer and the brush. In this regime, the COF decreases rapidly with an increase in So. Finally, in the hydrodynamic lubrication regime, the thickness of the fluid layer is larger than that of the roughness of the brush surface and there is hardly any contact between the wafer and the brush. The COF remains low and almost constant with increase in So in this regime.

Figure 4.7 shows the Stribeck curves at various applied pressures (0.25-0.58 psi) and brush speed (10-60 rpm) at different solution pH (1.1, 7.0, and 10.7) [34]. At an acidic pH of 1.1, the COF increases with increasing value of applied pressure for all brush speeds. At a fixed applied pressure, the COF decreases as the brush speed increases for pressure values in the range of 0.35-0.58 psi, indicating the presence of partial lubrication regime. However, at lower pressure of 0.25 psi, the COF remains almost constant with increasing brush speed suggesting the transformation of the regime from partial lubrication to hydrodynamic lubrication under this condition.

At neutral pH of 7.0, the COF again is higher for higher applied pressures for almost all values of So as was the case with acidic pH. However, the COF values are almost two times lower than their counterparts at low pH. Interestingly, when the pressure is kept constant, the COF does not change significantly with increasing brush speed, suggesting that the transition of tribological mechanism from partial to hydrodynamic lubrication. When the solution pH (10.7) is alkaline, the COF values are further lower than their counterparts at neutral and acidic pH. The COF values remain constant with increasing So at all pressures confirming that the system is maintained in the hydrodynamic lubrication regime.

The higher COF under acidic condition was explained based on the solubility and gelation characteristics of silica in silica-water system. At pH close to the ISE point of silica (pH $\sim$2), the silicic acid dissolves and aggregates into chains forming a three-dimensional network resulting in higher values of COF. The lower COF at neutral and alkaline pH was attributed to formation of lubricating network terminated silanol groups and soluble silicates, respectively.

An optimum level of shear force is needed between the brush and the wafer as low level of shear force will not remove particles, while excessive force will lead to scratches and other types of defects. Higher downforce has been shown to promote particle removal but also generates scratches [37]. Increasing the brush rotation rate at lower down pressure may offer suitable contact characteristic conditions for effective particle removal and lower defects due to increased contribution of the fluid flow. YaTing et al. highlighted the effect of brush rotation speed on removal of 500 nm polystyrene particles from glass samples with

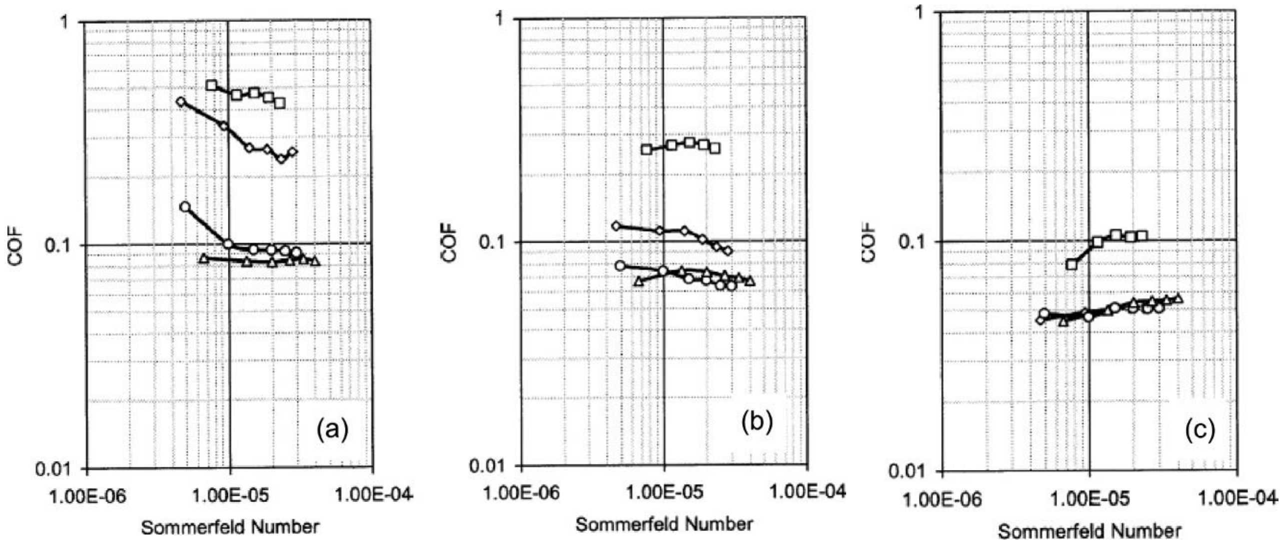

FIGURE 4.7 Effect of applied pressure and brush velocity on COF at fixed flow rate of the cleaning solution (60 mL/min): (□) 0.58, (◊) 0.45, (o) 0.35, (△) 0.25 psi; (a) pH of 1.1, (b) pH of 7.0, and (c) pH of 10.7. *(Reprinted from Ref. [34].)*

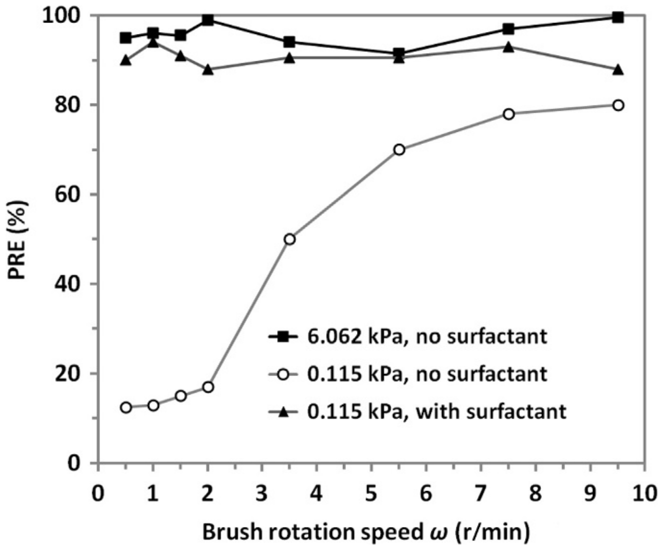

FIGURE 4.8 Effect of brush load, rotation speed, and addition of surfactants on particle removal-rate. *(Reprinted from Ref. [38].)*

copper film at applied pressures of 0.115 and 6.062 kPa and using DI water and 12% sodium dodecyl sulfate solution at a pH of 8 [38]. As may be noticed from Fig. 4.8, the effect of brush rotation speed is prominent only at lower applied pressures and when the cleaning solution is DI water. For brush speeds lower than 2 r/min, the particle removal efficiency (PRE) increases slowly with the speed and reaches a value of 20% at 2 r/min. Further increase in the brush speed between 2 and 5.5 r/min increases the PRE at a higher rate after which PRE appears to saturate at about 80%. The increase in PRE with brush rotation speed is due to higher hydrodynamic forces generated with increasing velocity gradients from higher speeds. Adding the surfactant (decreasing adhesion forces) or increasing the applied pressure (higher contact areas and friction between the substrate and the brush) increased the PRE to ≥90%. The authors concluded that in brush cleaning, it is easier to improve the PRE by increasing the brush load or by adding surfactant than by increasing the brush rotation speed. However, higher brush load may also lead to greater defects.

4 MEGASONIC CLEANING

The importance of cleaning processes that remove nanoparticles from polished surfaces without causing damage is continuously growing. The restrictions on material loss during cleaning prevent the use of aggressive chemicals. Dilute chemistries are used in combination with physical forces to drag the small particles on the surface [39]. Megasonic cleaning, a noncontact mode of cleaning, is commonly used as a means to apply such physical force to surfaces. This technique was first patented by Shwartzman, Mayer, and Kern of RCA Solid State Technology Center [40].

Generally, acoustic waves at MHz frequency range are generated by exciting a piezoelectric crystal made of ceramic. Piezoelectric crystals vibrate when they are subjected to an alternating electric field. When under optimized physical parameters and excited at its resonance frequency, most of the electrical energy can be converted into acoustic energy. The acoustic wave in the form of pressure fluctuations is propagated through a liquid medium. During acoustic propagation in liquids, two types of phenomena are known to take place, namely, acoustic streaming and acoustic cavitation. These two phenomena are considered to be the main mechanisms for particle removal. Acoustic streaming refers to time-independent liquid motion due to viscous attenuation, which is mainly classified into Rayleigh streaming, Eckart streaming, and Schlichting streaming. Eckart streaming typically occurs in a free nonuniform sound field, which is characterized by vortices much larger than the wavelength of sound [41], whereas Schlichting and Rayleigh streamings result in liquid flow close to a solid boundary with length scales much smaller or on the order of sound wavelengths [42]. The most important aspect of megasonic cleaning is the reduction of boundary layer thickness. Eckart streaming mainly reduces the diffusion boundary layer thickness and thus enhances the chemical reactivity at the interface. Schlichting streaming, also called boundary layer streaming, enhances the particle removal within the viscous boundary layer by a rolling mechanism.

The acoustic boundary layer thickness (δ) in a fluid with a kinematic viscosity of υ decreases with an increase in the angular acoustic frequency (ω) [43]

$$\delta = \left(\frac{2\upsilon}{\omega}\right)^{1/2} \tag{16}$$

At a frequency of 1 MHz in a sound field, the acoustic boundary layer thickness is approximately 0.5 μm. In comparison, the turbulent hydrodynamic boundary layer on a flat surface is greater than 1600 μm for a free stream velocity of 10 m/s in water.

Another important aspect in sound field is the acoustic streaming velocity (U_a), which is proportional to the product of the square of the acoustic frequency (ω) and transducer power density and inversely proportional to the viscosity (μ) [43]

$$U_a \propto \frac{\omega^2 I}{\mu} \tag{17}$$

where I is the transducer power density. For example, in an acoustic field at a frequency of ~1 MHz, if the transducer power density is 10 W/cm^2, the streaming velocity can be as high as 5.4 m/s in water. Thus, particles are exposed to a

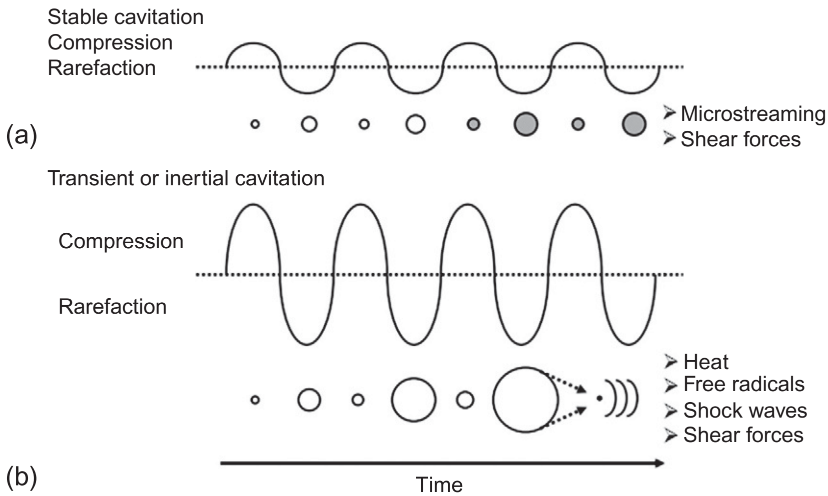

FIGURE 4.9 Schematic diagram of (a) stable cavitation and (b) transient cavitation. *(Reprinted from Ref. [44].)*

much higher drag than that in a hydrodynamic flow due to the viscous stresses and large velocity gradients combined with boundary layer thickness reduction, which enhances particle removal from a substrate.

Acoustic cavitation is the phenomenon that refers to the generation and motion of bubbles in liquid under acoustic irradiation. Bubbles are usually formed during the low-pressure cycle of the sound wave. When bubbles grow to an active size, they either continuously oscillate over many cycles (stable cavitation) or grow in size and eventually collapse in less than a few cycles (transient cavitation). Figure 4.9 shows a schematic diagram of the two types of cavitation [44].

Several effects are caused by cavitation events in an acoustic field. Stable cavitation occurs when bubbles oscillate around a resonant size. At 1 MHz, the resonant bubble size is roughly 3.7 μm. These bubbles act as secondary sources of sound and generate local shear forces and liquid flow adjacent to the bubble wall, a phenomenon called microstreaming. Stable cavitation will usually eventually disappear. Transient cavitation is characterized by large bubble size variations and eventual collapse. The dramatic compression forces involved in bubble collapse process can cause localized extremely high temperature (>4000 °C) and pressure conditions, which in turn are accompanied by secondary phenomena such as shock waves and microjets as well as free radical generation [45]. An important phenomenon used to indicate the level of transient cavitation is called sonoluminescence, which refers to light emission from inside of bubbles when excited free radicals fall back to their ground state. Several studies can be found on the effect of solution parameters and sound field variables on controlling the level of sonoluminescence signal [46–49]. Both

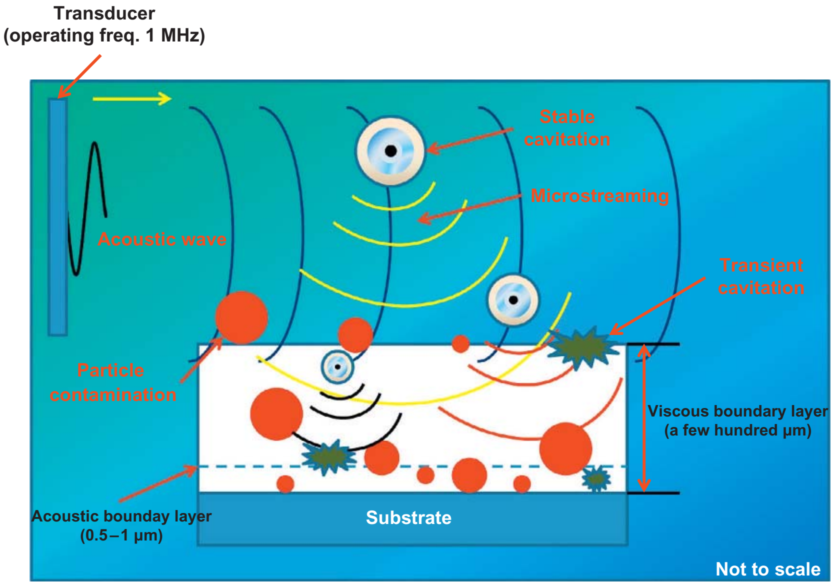

FIGURE 4.10 Schematic of different phenomena active in an acoustic field at a frequency of ~1 MHz.

microstreaming and shock waves or microjets are considered to be instrumental in particle removal [50,51]. Figure 4.10 presents some of the particle removal mechanisms that occur during the megasonic cleaning process.

Moumen et al. showed the effect of key parameters such as megasonic power, cleaning time, and solution temperature on cleaning using dilute SC-1 solutions in a post-CMP cleaning process [52]. They found that low defect count was accomplished in the cleaning of polished thermal oxide wafer under optimum power, temperature, and cleaning time. Additionally, a study of particle removal mechanisms in post-CMP cleaning of thermal oxide wafers using megasonics was conducted by the same authors in 2002 [53]. Thermal oxide wafers were polished using silica-based slurry (SC-112 by Cabot) and cleaned in a megasonic tank at a frequency of 850 ± 10-15 kHz. The ratio of components in SC-1 solution used in their study was H_2O:H_2O_2:NH_4OH = 40:2:1. The input megasonic power was varied from 100 to 500 W. Megasonic cleaning with SC-1 was proved to be a very effective for post-CMP cleaning. The results showed that megasonic power had no major effect on cleaning efficiency for cleaning time of 8 min or longer (as indicated from data shown in Fig. 4.11), suggesting that the acoustic powers used in their study were above the threshold limit required for achieving almost complete particle removal.

However, the SC-1 solution temperature had a major impact on cleaning efficiency and defect level. Figure 4.12 shows the results at different solution temperatures. When the temperature was increased from 35 to 55 °C, the

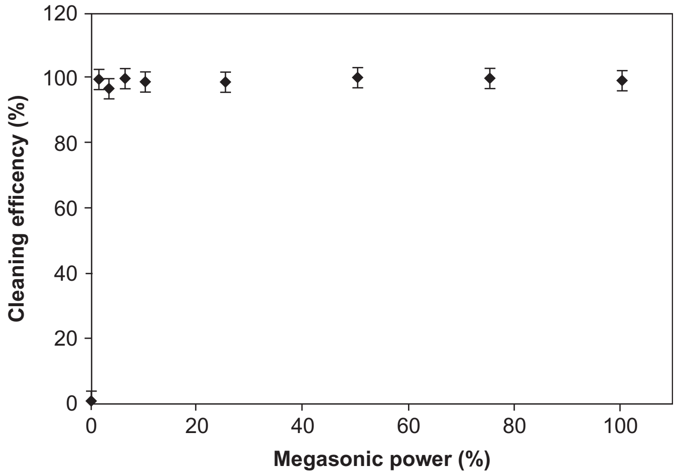

FIGURE 4.11 Cleaning efficiency versus megasonic power for SC-1 treatment at 8 min and 35 °C. *(Reprinted from Ref. [53].)*

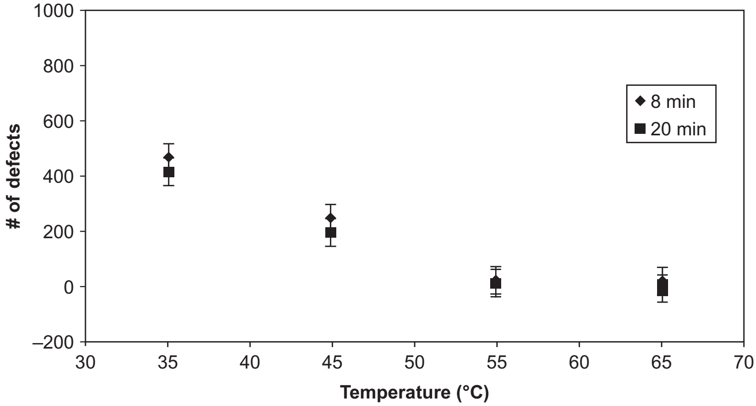

FIGURE 4.12 Effect of SC-1 solution temperature on number of defects. *(Reprinted from Ref. [53].)*

number of defects decreased dramatically; however, further increase in temperature above 55 °C did not change the defect numbers by much. The authors claimed that such an effect was due to an increase of stable cavitation activity with an increase in temperature from 23 to 65 °C. Effective cleaning was accomplished with the combined effect of stable cavitation and some etching.

In addition to the effect of temperature on the number of defects, the etching of oxide under megasonic irradiation was also measured. Figure 4.13 shows the effect of temperature on thickness of oxide etched under a given megasonic power level for different etching times. Clearly, below 45 °C, the etch rate

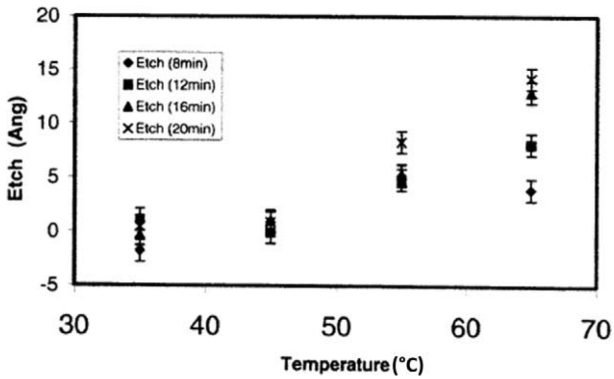

FIGURE 4.13 Effect of SC-1 solution temperature on etch rate of thermal oxide at a specified megasonic power. *(Reprinted from Ref. [53].)*

(oxide thickness etched divided by etching time) did not change regardless of etching time. With temperatures greater than 45 °C, the etch rate increased dramatically. However, etching alone from SC-1 solution was not sufficient to remove any particles. Reduction of defect number was due to the high etching rate of SC-1 solution at higher temperatures combined with stable cavitation generated during megasonic cleaning. The analysis of particle removal at very low megasonic power pointed out that the cleaning mechanism was the reduction of total adhesion force due to an increase of the electrostatic repulsion.

Huang et al. showed the advantage of using a combination of ultrasonic and megasonic fields on post-CMP cleaning of hard disk substrates [54]. The cleaning procedure used in their study was: ultrasonic cleaning at a frequency of 130 kHz after 15 min soaking, brush (made of poly(vinyl alcohol)) scrubbing using a detergent at a concentration of 0.5%, and 5 min ultrasonic cleaning at 170 kHz followed by 5 min megasonic cleaning at 960 kHz. The counts of different defect sizes were dramatically reduced when megasonic field was introduced (Fig. 4.14). Such results reveal the importance of megasonic field for particle removal at submicrometer level.

Koos et al. summarized and compared megasonic cleaning with brush cleaning employing different chemistries [55]. In their study, blanket copper film stack consisting of 1.5-μm copper film electrochemically deposited on a PVD TaN/Cu barrier/seed layer was used. The underlying dielectric was plasma-enhanced CVD TEOS-based silicon dioxide. Prior to using megasonic cleaning, the copper film was partially polished followed by cleaning using a chemical formulation at low pH. A unique chemistry was used in megasonic cleaning process performed on a single-wafer cleaning tool. Compared to traditional brush cleaning process under similar conditions and chemistries, megasonic cleaning reduced the defect level by ~50% on 300-mm copper wafer. Additionally, lower copper surface roughness was observed when using megasonic cleaning than

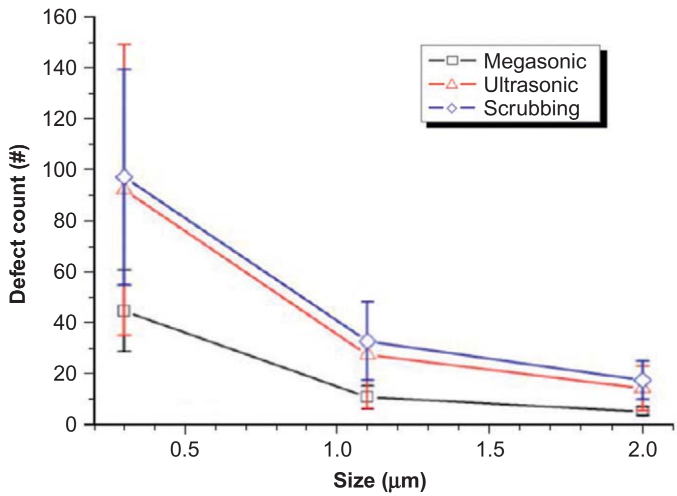

FIGURE 4.14 Defect count after each cleaning process. *(Reprinted from Ref. [54].)*

brush scrubbing. The effect of megasonic power indicated that in order to achieve more than 90% particle removal, power density of 2.3 W/cm^2 or higher was required. Further, contamination analysis on the bevel part of the copper wafers showed a clean surface after megasonic cleaning, which manifested the benefit of megasonic cleaning as a noncontact post-CMP copper cleaning method with the ability to clean surfaces as well as the edges.

5 CLEANING CHEMISTRIES

5.1 Silicon Dioxide Post-CMP Cleaning

In most oxide CMP processes, slurry containing silica abrasives is commonly used, which leaves gross particle contamination. Different chemistries are used to remove such particle contamination with different cleaning methods. For brush cleaning, the chemistry commonly used is NH_4OH-based solutions with citric acid and HF added. The concentration of NH_4OH is usually 1-2% mainly to remove particles and prevent redeposition. Citric acid (0.5%) is added for metal removal and HF etches oxide to remove subsurface defects. In the case of megasonic cleaning, chemicals used mostly are tetramethyl ammonium hydroxide (TMAH)-based solutions and SC-1.

Moumen et al. tested different variables that affected post-CMP particle removal on thermal oxide wafers using both contact and noncontact techniques [56]. The megasonic cleaning chemistry used in their study was SC-1 solution with a ratio of H_2O:H_2O_2:NH_4OH = 20:1:0.5. Particle removal was evaluated for megasonic and brush cleaning processes. In megasonic cleaning process, acoustic power, cleaning time, and SC-1 solution temperature were investigated for their effect on particle removal and level of defects. In brush cleaning with DI water, the effect of brush pressure, brush rotational speed, and cleaning time was examined. Defects were counted (particles above 200 nm size) on wafer surface after cleaning. Lower power, longer time, and higher temperature were shown to provide better PRE during megasonic cleaning. In brush cleaning process, a higher rotation speed and pressure and a shorter cleaning time were required to achieve good particle removal. Figures 4.15 and 4.16 show the number of defects after cleaning using a megasonic field in SC-1 solution and brush scrubbing using DI water only.

Two different megasonic powers were used, namely, 150 and 540 W with SC-1 solutions at temperatures of 30 and 60 °C for cleaning times of 5 and

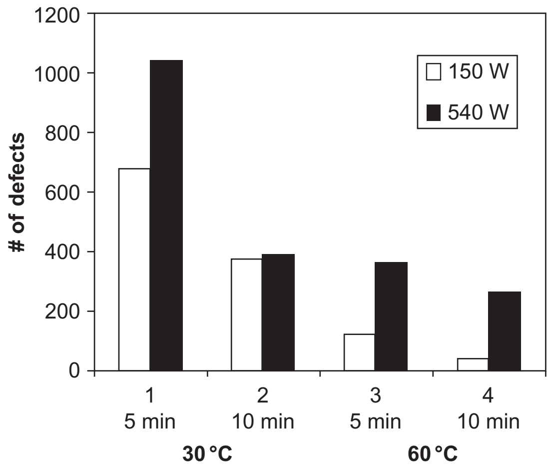

FIGURE 4.15 Defect level after megasonic cleaning. *(Reprinted from Ref. [56].)*

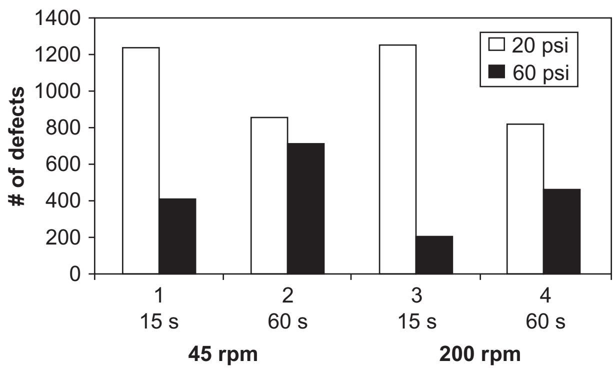

FIGURE 4.16 Defect level after brush cleaning. *(Reprinted from Ref. [56].)*

10 min. The lowest defect number, as shown in Fig. 4.15, was measured for megasonic power of 150 W at 60 °C for 10 min of cleaning. In comparison, the lowest defect number for brush cleaning was achieved under a brush pressure of 60 psi at a rotation speed of 200 rpm for 15 s cleaning using DI water. The results indicated comparable cleaning performance using the two techniques; however, noncontact cleaning using SC-1 showed lower defect counts. The study revealed that using DI water in brush cleaning, cleaning performance comparable to megasonic cleaning can be achieved without any use of chemistry.

A study on the effect of particle aging on removal efficiency was conducted by Busnaina and Park where silicon dioxide wafers were dipped into ceria slurry and spun or air dried [57]. Cleaning was conducted using SC-1 or DI water in the presence of megasonic field. The results are shown in Fig. 4.17. The higher the aging time, the lower was the PRE. Aging increases the contact area between the particles and therefore the adhesion force. In the case of air dry, significant capillary forces may be present, which can dramatically increase the adhesion of particles to the substrate possibly due to hydrogen or covalent bond formation. Interestingly, SC-1 aided by megasonic field for the wafers that were spun dried after deposition exhibited highest removal efficiency.

In addition to particle removal, metallic contamination and the dielectric damaged layer (caused by mechanical effects of CMP) are also needed to be removed during post-CMP cleaning process. Tardif et al. optimized the cleaning process after oxide CMP process for particle removal, damaged oxide layer, and metallic trace removal [58]. Different wet chemistries were tested for SiO_2 slurry particle removal from plasma-enhanced chemical vapor deposited oxide

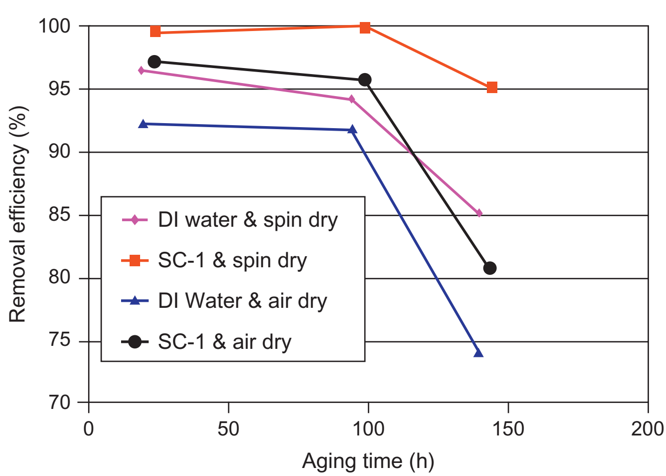

FIGURE 4.17 Removal efficiency of aged ceria particles on SiO_2 wafers. *(Reprinted from Ref. [57].)*

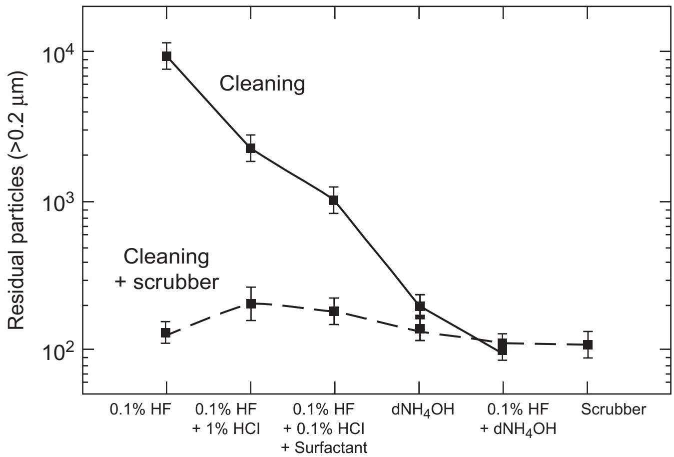

FIGURE 4.18 Residual particles for scrubbing and different wet chemistries. *(Reprinted from Ref. [58].)*

films from tetraethylorthosilicate. The cleaning time was chosen to be 10 min and cleaning was conducted with and without scrubbing in different solutions. The numbers of residual particles at the end of the process are plotted for different chemical treatments in Fig. 4.18. DHF solutions (0.1%) with 1% HCl provided better cleaning than HF solution alone. Addition of a surfactant further reduced the number of residual particles. However, the best cleaning results were observed with HF (0.1%) dip for 1 min followed by dilute ammonium hydroxide treatment. Using the scrubber in conjunction with the wet chemistries offered superior cleaning in almost all cases.

For damaged dielectric layer removal, diluted HF or hot alkaline bath (NH_4OH) was used as the preferred solution. The authors illustrated a determination of damaged layer defined from the oxide etch rate change using the two solutions (Fig. 4.19). In the particular example shown in the figure, the damaged oxide layer thickness was estimated to be about 3 nm.

The metal contaminants deposited were mainly Al and Fe from alkaline baths. Hydrofluoric acid and mixtures of hydrofluoric acid and hydrochloric acid solutions can be used for noble metals where the oxide does not contain nonoxidized silicon atoms. The authors claimed that the most optimal cleaning after silicon oxide CMP consists of a first megasonic SC-1 (with or without H_2O_2) for both particle and damaged layer removal followed by a 0.5 min HF/HCl dip to remove the metal traces.

Liu et al. investigated the mechanism of particle removal from thermally grown silicon dioxide and chemical vapor deposited silicon nitride films on silicon wafers by immersion in aqueous solutions of various pH for 30 min [2]. The films were contaminated with fused silica particles during polishing using

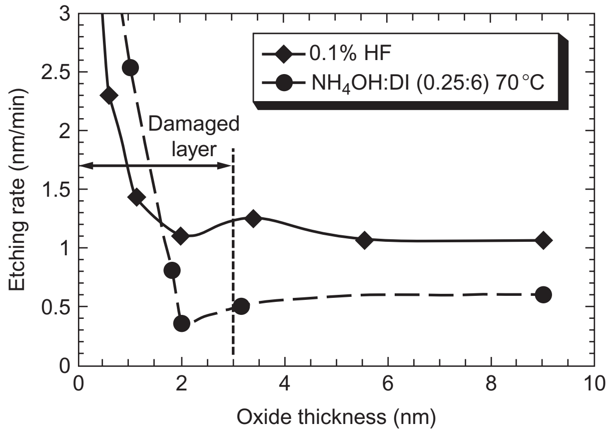

FIGURE 4.19 Example of damaged layer estimation. *(Reprinted from Ref. [58].)*

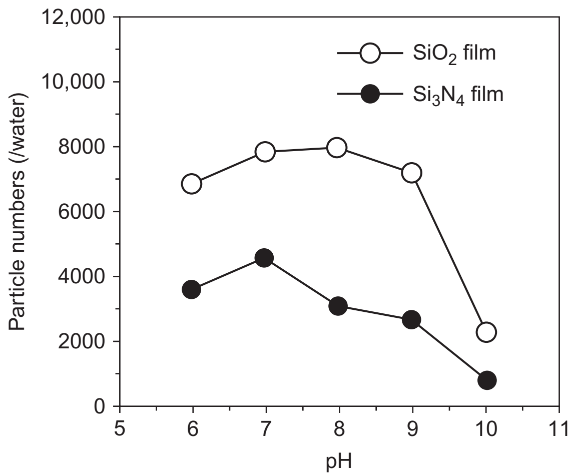

FIGURE 4.20 Particle numbers on silicon dioxide and silicon nitride films immersed in various pH solutions for 30 min after CMP process. *(Reprinted from Ref. [2].)*

commercially available slurry. Their results displayed in Fig. 4.20 indicated that the particle counts reduced as the pH was increased in the alkaline range. This effect was explained based on the dissolution of silica primarily as $Si(OH)_4$ for all pH values ≥ 8 and increasing magnitude of zeta potential with pH, which generates higher electrostatic repulsive forces between the particle and the surface. The higher particle count in the case of silica film was attributed to lower hardness (36 GPa) of silica compared to silicon nitride (50 GPa) resulting in greater adhesion and embedment of particles in softer films.

5.2 Tungsten Post-CMP Cleaning

For tungsten (W) post-CMP cleaning, alumina particles are commonly present as contaminants due to the use of these particles as abrasives in the polishing slurry [5]. Particles trapped inside tungsten plug reduce the reliability of electronic devices by blocking the contact between the W plug and the upper metal line [3]. As a weak reducing agent with $E_{0W/WO_3} = -0.09$ V, W can be oxidized into WO_2, W_2O_5, or WO_3. Corrosion is usually very low in ammonia solutions when WO_3 is formed. WO_3 is also stable at pH lower than 4. In the process of brush cleaning for post-W CMP, particles are usually difficult to remove in the recessed plugs since the brushes are not able to reach them easily [3]. DHF has been used in brush cleaning process, which forms fluorinated tungsten complexes. However, not all of the devices are allowed to be exposed to HF chemistry due to their poor electrical characteristics [59].

Kang et al. investigated the effect of W plug geometry on PRE using ammonia solutions [60]. Their experiments were conducted using a poly(vinyl alcohol) (PVA) brush scrubber. Figure 4.21 shows the number of slurry particles left on recessed, flat, and protrusion W plugs. The authors claimed that the poor removal of silica particles on recessed W plugs was due to the ineffective transfer of brush removal force to the particles. Figure 4.22 shows scanning electron microscope images of the three W plug geometries.

Based on the different particle removal performances with three geometries of W plugs, the authors proposed modification of W plug geometry by buffing the W plug with oxide slurry. Using oxide slurry, which has high selectivity between oxide and W, faster removal of oxide than W resulted in the protruded geometry of W plug. In this case, the removal forces were transferred to the particles effectively and no particles were observed after post-CMP cleaning using ammonia solutions.

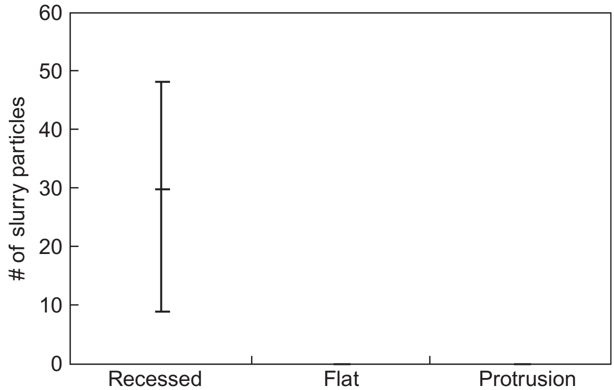

FIGURE 4.21 Number of slurry particles left on recessed, flat and protrusion W plugs. *(Reprinted from Ref. [60].)*

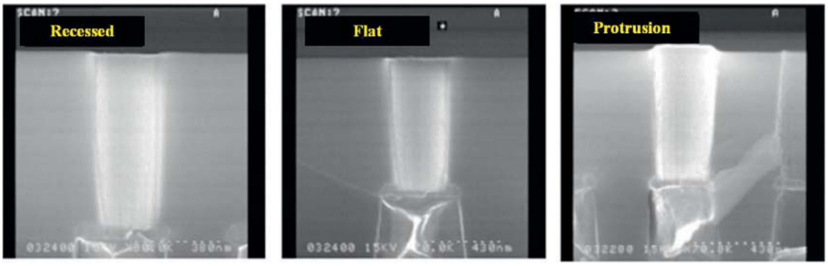

FIGURE 4.22 SEM images of recessed, flat and protrusion geometries of W plugs. *(Reprinted from Ref. [60].)*

TABLE 4.7 Post-W CMP Cleaning Recipe

Step No.	Step Name	Item	Old Conditions	Conditions Optimized
1	Mega tank	Filter	None	Assembled
		Replenish time	4-7 h	0.25-0.4 h
		NH_4OH	0.09 wt.%	2 wt.%
2	Brush 1	NH_4OH	2 wt.%	2 wt.%
		Brush	Open	Closed
3	Brush 2	HF %	0.5 wt.%	0.5 wt.%
		Brush	Closed	Open
4	SRD	-	Without any change	

SRD, spin rinse dry.
Reprinted from Ref. [61].

Cheng optimized post-W CMP cleaning process by combining megasonic and brush cleaning methods employing ammonia and HF solutions [61]. Table 4.7 shows the cleaning recipe used in his study.

For sub-130 nm logic product, yield loss has been greatly reduced by decreasing defect numbers after post-W CMP cleaning by optimization of combined use of megasonic and brush cleaning.

5.3 Copper Post-CMP Cleaning

Copper is the material of choice for interconnects in today's IC manufacturing due to its lower resistivity and higher resistance to electromigration compared to aluminum. Copper interconnects are formed using a dual damascene process at the end of which CMP is used to remove excess copper and Ta/TaN adhesion/barrier layer. Colloidal silica, the most commonly used abrasive in copper CMP slurry, chemisorbs onto the copper oxide (Cu_2O or CuO) and copper hydroxide ($Cu(OH)_2$) formed through oxidation of copper by hydrogen peroxide in the slurry [62]. Chen et al. developed a cleaning method for the removal of colloidal silica abrasive from polished copper surface using a two-step process [62]. The first step involved a buffing step employing dilute HNO_3/benzotriazole (BTA) aqueous solution that causes etching of the copper oxide and releases the attached silica particles by breaking the chemical bond between silica and the copper oxide. Figure 4.23 illustrates bridging of silica abrasive to copper hydroxide by means of oxygen and removal of the particle using the buffing step. The copper surface is passivated with BTA at the end of this step. The optimum concentrations and buffing time that yielded lowest dishing of ~5 nm

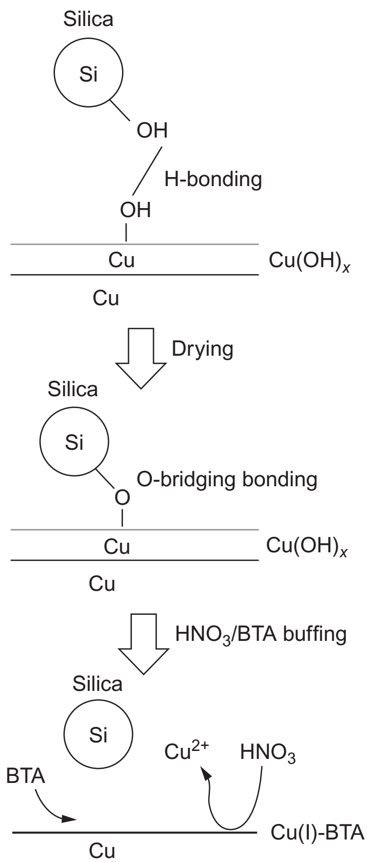

FIGURE 4.23 Schematic representation of bridging of colloidal silica abrasive to copper hydroxide and the removal of particle using an aqueous solution of HNO_3 and BTA. *(Reprinted from Ref. [62].)*

were identified to be 3 vol.% HNO_3 and 5 mM BTA and 10 s, respectively. In order to prevent redeposition of silica abrasive on a hydrophobic Cu-BTA surface, a second PVA brush scrubbing step using a nonionic surfactant, Triton X-100, was shown to be critical. It was hypothesized that adsorption of surfactant molecules at critical micelle concentration level renders the Cu-BTA surface hydrophilic, which in conjunction with PVA brush scrubbing cleans any residual colloidal silica as well as prevents its redeposition by steric barrier mechanism.

Hong et al. determined the role of additives such as BTA and pH adjusters (ammonium hydroxide and TMAH) on adhesion and removal of silica particles to/from copper films in citric acid-based post-copper CMP cleaning solutions [63]. The zeta potential of both silica particles (30 nm) and copper increased in magnitude with addition of citric acid in 1 mM KCl solutions. The adsorption of citrates on these surfaces was explained as the reason for higher surface charge and the resulting high zeta potential. Adhesion studies were in line with zeta potential measurements and showed a decrease in the adhesion force between silica particles and copper with increasing citric acid concentration due to higher electrostatic repulsion between the surfaces.

The authors also calculated the theoretical total interaction force using the DLVO theory between silica particle and copper surface as a function of separation distance in various cleaning formulations. Their results, shown in Fig. 4.24, indicated that the strongest attractive force was obtained in the BTA and citric acid solution containing TMAH, while the weakest attractive interaction was calculated for the same solution but with TMAH replaced by NH_4OH. The adhesion force was also measured experimentally in these cleaning solutions and compared to that in DI water (Fig. 4.24). The smallest adhesion of 0.0124 nN was measured in citric acid solutions with pH adjusted to 6.0 using ammonium hydroxide, while largest adhesion force of 8.87 nN was measured in TMAH containing citric acid solutions of pH 6.0. The much higher adhesion in TMAH solutions was attributed to the adsorption of TMA ions on silica and copper causing these surfaces to exhibit lower zeta potential values as measured experimentally. The highest particle removal efficiency was measured for cleaning solutions that yielded lowest adhesion force, thereby highlighting the importance of pH adjuster in cleaning formulations.

Venkatesh et al. conducted polarization studies on copper treated with BTA for 1 min and subsequently with solutions containing TMAH only, TMAH with arginine (1%) and TMAH with arginine (1%) and uric acid (0.01 M) [64]. The role of TMAH was that of cleaning agent; arginine, a chelating agent for copper and uric acid, was proposed as replacement for BTA to inhibit copper corrosion. The importance of achieving a balance in the cleaning chemistry was highlighted so that minimal or no etching of copper occurs and any etched copper is complexed to prevent its redeposition. Figure 4.25 shows the corrosion current density estimated from the polarization curves for various cleaning formulations. The current density was lowest for DI water as no BTA was removed

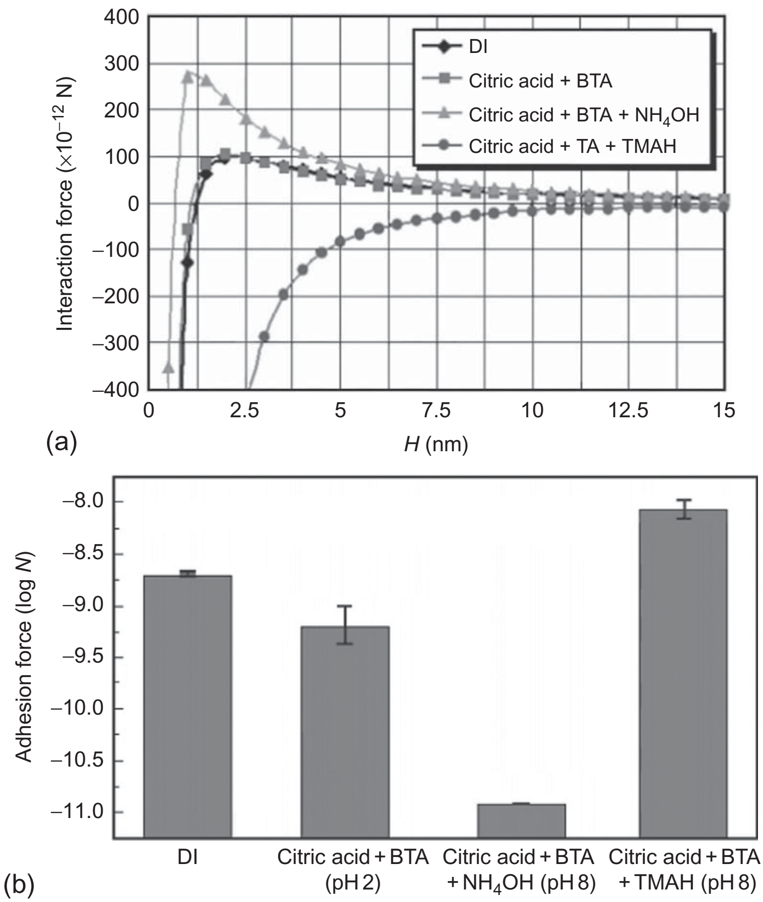

FIGURE 4.24 (a) Calculated interaction force between silica particles and copper surface using DLVO theory in various cleaning formulations, (b) adhesion forces between silica particles and copper surface in post-CMP Cu cleaning chemistries. *(Reprinted from Ref. [63].)*

in this case. TMAH alone was identified as the most favorable formulation for not only effectively removing BTA but also providing lowest copper corrosion. Uric acid was not found to offer any additional benefit in reducing copper corrosion rate.

Particle contamination and cleaning studies were performed by dipping copper samples in a slurry containing 12.5 wt.% silica abrasive of size 100 nm and treating the samples in various solutions. The field emission scanning electron microscopy (FE-SEM) images of fresh copper sample, copper sample treated with DI water, and solution of TMAH, arginine, and uric acid are shown in Fig. 4.26a through c, respectively. It is clear from these images that the treatment of DI water results in a significant number of particles remaining on the

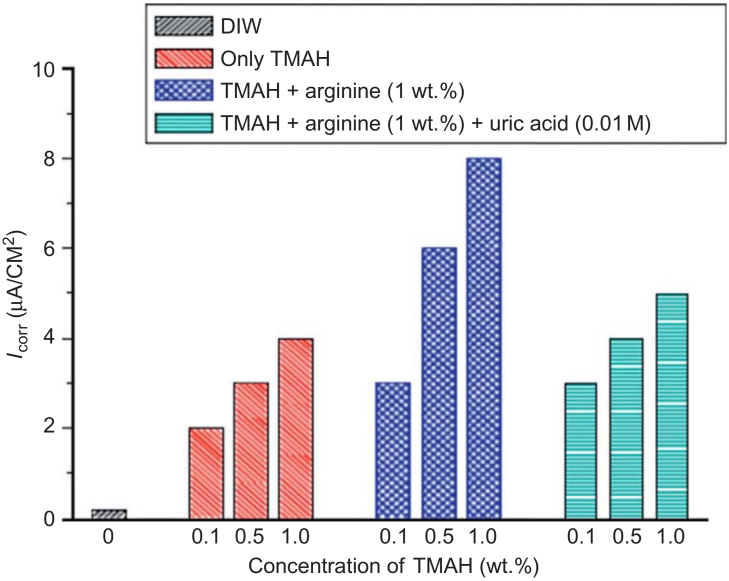

FIGURE 4.25 Corrosion current density of Cu in various solutions. *(Reprinted from [64].)*

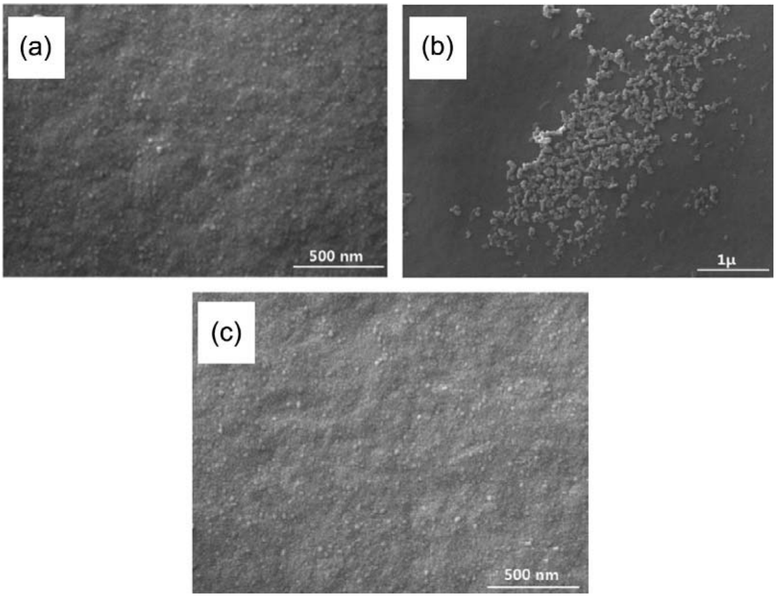

FIGURE 4.26 FE-SEM micrographs of various Cu samples (a) fresh Cu, (b) dipped in silica slurry and treated with water, (c) dipped in silica slurry and treated with solutions containing 0.5 wt.% TMAH, 1 wt.% arginine, and 0.01 M uric acid. *(Reprinted from Ref. [64].)*

surface. In the case of TMAH/arginine/uric acid treated copper sample, it appears very clean and the particles appear to be completely removed. The effectiveness of TMAH formulation in achieving good cleaning was attributed to the high pH of the solution causing the surface charge of the particle and substrate to be negative and the interactive forces to be repulsive. It was proposed that TMAH removed BTA film, facilitated the wetting of the surface (to make it hydrophobic), and enabled slight etching of the copper required for particle removal. Arginine complexed the etched copper ions and aided in preventing their redeposition.

6 SUMMARY

Post-CMP cleaning is a critical step after CMP as it achieves effective removal of particulate and other contaminants for improvement of device yield in IC fabrication. The studies described in this chapter emphasize the importance of control of process variables in single wafer (brush and megasonic cleaning) and immersion cleaning for enabling high PRE without inducing any defects on the surface. The role of different chemical formulations and the compositions used in post-CMP cleaning of silicon dioxide, tungsten, and copper films in particle removal has been highlighted through discussion of various studies. The developments in post-CMP process must continuously progress in order to meet the emerging challenges and stricter requirements of surface cleaning at the end of CMP.

REFERENCES

[1] P.B. Zantye, A. Kumar, A.K. Sikder, Chemical mechanical planarization for microelectronics applications, Mater. Sci. Eng. R 45 (2004) 89.
[2] C. Liu, B. Dai, C. Yeh, Post cleaning of chemical mechanical polishing process, Appl. Surf. Sci. 92 (1996) 176.
[3] J.-G. Park, A.A. Busnaina, Y.-K. Hong, Post-CMP cleaning, in: Y. Li (Ed.), Microelectronic Applications of Chemical Mechanical Planarization, John Wiley & Sons, Hoboken, NJ, 2008, pp. 467–505.
[4] L. Zhang, S. Raghavan, M. Weling, Minimization of chemical–mechanical planarization (CMP) defects and post-CMP cleaning, J. Vac. Sci. Technol. B 17 (1999) 2248.
[5] F. Tardif, Post-CMP clean, in: S.H. Li, R. Miller (Eds.), Chemical Mechanical Polishing in Silicon Polishing, Semiconductors and Semimetals, vol. 63, Academic Press, London, U.K, 2000, pp. 183–214.
[6] P.C. Hiemenz, R. Rajagopalan, Principles of Colloid and Surface Chemistry, CRC Press, Boca Raton, FL, 1997.
[7] G. Kumar, S. Beaudoin, Undercut removal of micrometer-scale particles from surfaces, J. Electrochem. Soc. 153 (2006) G175.
[8] B. Derjaguin, Y. Rabinovich, N. Churaev, Direct measurement of molecular forces, Nature 272 (1978) 313.

[9] H. Ackler, R. French, Y. Chiang, Comparisons of Hamaker constant for ceramic systems with intervening vacuum or water: From force laws and physical properties, J. Colloid Interface Sci. 179 (1996) 460.

[10] K. Johnson, K. Kendall, D. Roberts, Surface energy and the contact of elastic solids, Proc. Royal Soc. London A 324 (1971) 301.

[11] D. Maugis, H. Pollock, Surface forces, deformation and adherence at metal microcontacts, Acta Metall. 32 (1984) 1323.

[12] B. Derjaguin, V. Muller, Y. Toporov, Effect of contact deformations on the adhesion of particles, J. Colloid Interface Sci. 53 (1975) 314.

[13] H. Zhou, M. Gotzinger, W. Peukert, The influence of particle charge and roughness on particle-substrate adhesion, Powder Technol. 135–136 (2003) 82.

[14] Y. Rabinovich, J. Adler, A. Ata, R. Singh, B. Moudgil, Adhesion between nanoscale rough surfaces, J. Colloid Interface Sci. 232 (2000) 10.

[15] M.B. Ranade, Adhesion and removal of fine particles on surfaces, Aerosol Sci. Technol. 7 (1987) 161.

[16] W. Kern, The evolution of silicon wafer cleaning technology, J. Electrochem. Soc. 137 (1990) 1887.

[17] J. Ruzyllo, Wet Cleaning Technology-Implementation, Semiconductor Note 17, Posted April 19 (2006). http://www.semi1source.com/notes/ViewFile.asp?Which=72.

[18] H. Abe, H. Iwamoto, T. Toshima, T. Iino, G.W. Gale, Novel photoresist stripping using ozone/vaporized water mixture, IEEE Trans. Semicond. Manuf. 16 (2003) 401.

[19] H. Tomita, M. Sato, S. Nadahara, T. Saitoh, Photoresist stripping using novel sulfuric/ozone process, in: Proc. 2001 IEEE Intl. Symp. on Semiconductor Manufacturing, IEEE, Piscataway, NJ, 2001, pp. 199–202.

[20] S. Raghavan, M. Keswani, Thermal and chemical characteristics of SPM systems for resist stripping applications, in: SEMI Technology Symposium (STS), vol. 2, SEMICON Korea Seoul, Korea, 2008, pp. 223–227.

[21] G.A. Olah, Á. Molnár, Hydrocarbon Chemistry, second ed., Wiley-InterScience, New York, NY, 2003.

[22] X.G. Zhang, Electrochemistry of Silicon and Its Oxide, Kluwer Academic/Plenum Publishers, New York, NY, 2001.

[23] W. Kern, C.A. Deckert, Chemical etching, in: J.L. Vossen, W. Kern (Eds.), Thin Film Processes, Academic Press, New York, NY, 1978, pp. 401–496, Chapter V-1.

[24] W. Kern, G. Schnable, Wet etching, in: S.J. Moss, A. Ledwith (Eds.), Chemistry of the Semiconductor Industry, Chapman and Hall, New York, NY, 1987, pp. 223–276, Chapter 11.

[25] A. Abbadie, F. Crescente, M. Semeria, Advanced wet cleanings post-CMP: application to reclaim wafers, J. Electrochem. Soc. 151 (2004) G57.

[26] W. Tseng, V. Devarapalli, J. Steffes, A. Ticknor, M. Khojasteh, P. Poloju, C. Goyette, D. Steber, L. Tai, S. Molis, M. Zaitz, E. Rill, M. Kennett, L. Economikos, N. Lustig, C. Bunke, C. Truond, M. Chudzik, S. Grunow, Post copper CMP hybrid clean process for advanced BEOL technology, IEEE Trans. Semicond. Manuf. 26 (2003) 493.

[27] R. Chein, W. Liao, Modeling of particle removal using non-contact brush scrubbing in post-CMP cleaning processes, J. Adhesion 82 (2006) 555.

[28] R.A. Bowling, An analysis of particle adhesion on semiconductor surfaces, J. Electrochem. Soc. 132 (1985) 2208.

[29] L. Talbot, R.K. Cheng, R.W. Schefer, D.R. Willis, Thermophoresis of particles in a heated boundary layer, J. Fluid Mech. 101 (1980) 737.

[30] G. Zhang, G. Burdick, F. Dai, T. Bibby, S. Beaudoin, Assessment of post-CMP cleaning mechanisms using statistically designed experiments, Thin Solid Films 332 (1998) 379.
[31] K. Xu, R. Vos, G. Vereecke, G. Doumen, W. Fyen, P. Mertens, M. Heyns, C. Vinckier, J. Fransaer, Particle adhesion and removal mechanisms during brush scrubber cleaning, J. Vac. Sci. Technol. B 22 (2004) 2844.
[32] S.R. Roy, I. Ali, G. Shinn, N. Furusawa, R. Shah, S. Peterman, K. Witt, S. Eastman, P. Kumar, Postchemical-mechanical planarization cleanup process for interlayer dielectric films, J. Electrochem. Soc. 142 (1995) 216.
[33] A.A. Busnaina, H. Lin, N. Moumen, J.-W. Feng, J. Taylor, Particle adhesion and removal mechanisms in post-CMP cleaning processes, IEEE Trans. Semicond. Manuf. 15 (2002) 374.
[34] A. Philipossian, L. Mustapha, Tribological attributes of post-CMP brush scrubbing, J. Electrochem. Soc. 151 (2004) G456.
[35] A. Phillipossian, S. Olsen, Fundamental tribological and removal rate studies of inter-layer dielectric chemical mechanical planarization, Jpn. J. Appl. Phys. 42 (2003) 6371.
[36] K.C. Ludema, Friction, Wear, Lubrication: A Textbook in Tribology, CRC Press, Boca Raton, FL, 1996.
[37] X. Gu, T. Nemoto, A. Teramotb, T. Ito, S. Sugawa, T. Ohmi, Reduction of scratch on brush scrubbing in post CMP cleaning by analyzing contact kinetics on ultra low-k dielectric, ECS Trans. 19 (2009) 103.
[38] Y.T. Huang, Y. Li, D. Guo, C.L. Ming, Probing particle removal in brush scrubber cleaning with fluorescence cleaning, Sci. China Technol. Sci. 56 (2013) 2994.
[39] H. Kawahara, Removal of particulate contaminants, in: T. Hattori (Ed.), Ultraclean Surface Processing of Silicon Wafers. Secrets of VLSI Manufacturing, Springer-Verlag, Berlin and Heidelberg, Germany, 1998, pp. 451–461.
[40] S. Schwartzman, A. Mayer, W. Kern, Megasonic particle removal from solid state wafers, RCA Rev. 46 (1985) 81.
[41] G.W. Gale, A.A. Busnaina, Roles of cavitation and acoustic streaming in megasonic cleaning, Particulate Sci. Technol. 17 (1999) 229.
[42] H. Schlichting, K. Gersten, Boundary Layer Theory, eighth ed., Springer-Verlag, Berlin and Heidelberg, Germany, 2000.
[43] M. Keswani, Megasonic Cleaning of Wafers in Electrolyte Solutions: Possible Role of Electroacoustic and Cavitation Effects, Ph.D. Dissertation, University of Arizona, Tucson, AZ, 2008.
[44] C.M.H. Newman, T. Bettinger, Gene therapy progress and prospects: ultrasound for gene transfer, Gene Ther. 14 (2007) 465.
[45] Z. Han, M. Keswani, E. Liebscher, M. Beck, S. Raghavan, Analysis of sonoluminescence signal from megasonic irradiated gas-containing aqueous solutions using replaceable single-band filters, ECS J. Solid State Sci. Technol. 3 (2014) N3101.
[46] S. Helbig, S. Urban, E. Klein, S. Singh, Impact of megasonic process conditions on pre and sub-resolution assist feature damage, in: H. Kawahira, L.S. Zurbrick (Eds.), Photomask Technology, vol. 7122, SPIE, Bellingham, WA, 2008.
[47] K. Muralidharan, M. Keswani, H. Shende, P. Deymier, S. Raghavan, F. Eschbach, A. Sengupta, Experimental and simulation investigations of acoustic cavitation in megasonic cleaning, in: M.J. Lercel (Ed.), Emerging Lithographic Technologies XI, vol. 6517, SPIE, Bellingham, WA, 2007.
[48] F.R. Young, Sonoluminescence from water containing dissolved gases, J. Acoust. Soc. Am. 60 (1976) 100.

[49] Y.T. Didenko, T.V. Gordeychuk, V.L. Koretz, The effect of ultrasound power on water sonoluminescence, J. Sound Vib. 147 (1991) 409.

[50] G.W. Gale, A.A. Busnaina, Removal of particulate contaminants using ultrasonics and megasonics: a review, Particulate Sci. Technol. 13 (1995) 197.

[51] D. Chen, S.K. Sharma, A. Mudhoo (Eds.), Handbook on Applications of Ultrasound: Sonochemistry for Sustainability, CRC Press, Boca Raton, FL, 2011.

[52] N. Moumen, J. Piboontum, A.A. Busnaina, The effect of relative-humidity on particle adhesion and removal, in: Proc. 22nd Annual Meeting of The Adhesion Society, 1999.

[53] A.A. Busnaina, Fundamental cleaning mechanisms in post-CMP cleaning of thermal oxide wafers, in: Proc. 21st Annual Semiconductor Pure Water and Chemicals Conference (SPWCC), 2002, pp. 157–170.

[54] Y. Huang, X. Lu, G. Pan, B. Lee, J. Luo, Particles detection and analysis of hard disk substrate after cleaning of post chemical mechanical polishing, Appl. Surf. Sci. 255 (2009) 9100.

[55] D.A. Koos, J. Svirchevski, D.J. Vitkavage, D.G. Hansen, K.A. Reinhardt, F. Huang, M. Mitchel, G. Zhang, Development of a non-contact post-CMP cleaning process for copper, in: K.L. Mittal (Ed.), Particles on Surfaces 9: Detection, Adhesion and Removal, VSP, Leiden, The Netherlands, 2006, pp. 127–144.

[56] N. Moumen, M. Guarrera, C. Piboontum, A.A. Busnaina, Contact and non contact post-CMP cleaning of thermal oxide silicon wafers, in: Proc. 10th Annual IEEE/SEMI Advanced Semiconductor Manufacturing Conference and Workshop, 1999, pp. 250–253.

[57] A.A. Busnaina, J.-G. Park, The removal of particles in post-CMP applications, in: Proc. of the Planarization and CMP Technical Committee, The Japan Society for Precision Engineering, Fukuoka, Japan, 2003.

[58] F. Tardif, I. Constant, T. Lardin, O. Demolliens, M. Fayolle, Y. Gobil, J. Palleau, J. Torres, Cleaning after silicon oxide CMP, Microelectronic Eng. 37–38 (1997) 285.

[59] G.S. Cho, H.S. Kim, J.K. Lee, J.D. Jeong, D.Y. Kim, Device failure mechanism through particle adhesion, CMP-MIC 9 (2005) 263.

[60] Y.-J. Kang, C.-K. Yang, T.-Y. Kwon, J.-G. Park, J.-H. Jo, G.-S. Lim, Post W CMP cleaning without HF cleans, ECS Trans. 11 (2007) 441.

[61] X.-L. Cheng, Optimizing post cleaning of tungsten contact CMP to improve the yield of logic products with copper interconnect, in: Proc. 8th International Conference on Solid-State and Integrated Circuit Technology, ICSICT. IEEE, Piscataway, NJ, 2006, pp. 351–353.

[62] P.-L. Chen, J.-H. Chen, M.-S. Tsai, B.-T. Dai, C.-F. Yeh, Post-Cu CMP cleaning for colloidal silica abrasive removal, Microelectronic Eng. 75 (2004) 352.

[63] Y.-K. Hong, D.-H. Eom, S.-H. Lee, T.-G. Kim, J.-G. Park, A.A. Busnaina, The effect of additives in post-Cu CMP cleaning on particle adhesion and removal, J. Electrochem. Soc. 151 (2004) G756.

[64] R.P. Venkatesh, T.-Y. Kwon, Y.N. Prasad, S. Ramanathan, J.-G. Park, Characterization of TMAH based cleaning solution for post Cu-CMP application, Microelectronic Eng. 102 (2013) 74.